DELL Technologies

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Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dell Technologies Response to the US AI Action Plan RFI

A Blueprint for U.S. AI Global Leadership

Dell Technologies appreciates the opportunity to contribute to the United States AI Action Plan. As the technology industry's leading end-to-end AI infrastructure provider, and for more than 40 years a trusted partner and collaborator to the U.S. government, we look forward to continuing to help the administration strengthen America's global leadership in AI, drive government efficiency, create jobs and boost the U.S. economy. Our response is built on a three-pillar framework that can accelerate U.S. AI advancement: **Invest**—an infrastructure-first approach to AI innovation by prioritizing scalable, resilient systems; **Innovate**—driving AI leadership through public-private collaboration to pioneer transformative solutions; and **Evolve**—implementing agile AI policy and future-ready governance to lead the way in an ever-changing global landscape.

Realizing AI's full potential will require substantial and appropriate investment focus to help address chip resourcing and increased compute power, data storage and energy efficiency needs. Public and private sector collaboration provides the optimum basis from which to move forward, with the AI Action Plan now representing a generational opportunity to translate ambition into action, together.

Invest: An Infrastructure-First Approach to AI Innovation

1. Scaling AI Compute Infrastructure to Maintain Leadership

The U.S. should prioritize AI compute expansion to address escalating demand and prevent reliance on foreign supply chains. The rapid growth of generative AI and machine learning models necessitates significant advancements in AI infrastructure, high-performance computing (HPC), data center efficiency and semiconductor design and fabrication.

AI and data-driven decision making will play a central role in reshaping how governments operate, interact with citizens and provide essential services. We recommend the Trump administration continue to invest in and scale up compute infrastructure as a key priority to build a strong foundation for AI adoption and innovation across the nation. This infrastructure-first approach will ensure the U.S. remains competitive in the global AI landscape, secures its technological sovereignty, and unlocks accessible, transformative opportunities in both the public and private sectors. By addressing the growing compute demands and aligning infrastructure with strategic priorities, the administration can lay the groundwork for sustained leadership and economic prowess in AI innovation.

Key Considerations:

AI Compute Demand Growth:

- **AI Workloads**: According to IDC, genAI workloads will increase from 7.8% of the overall AI server market to 36% between 2022 and 2027. In storage, the curve is similar, with growth from 5.7% of AI storage from 2022 to 30.5% in 2027. Furthermore, Goldman Sachs research estimates that data center power consumption will grow by 160% by 2030, rising from 1–2% of global power usage to 3–4%. This demand calls for robust AI enabled infrastructure and compute power to meet the growing needs of AI.
- **Data Security:** Expanding AI compute infrastructure within the U.S. is essential to provide businesses and innovators with secure, domestic access to advanced computing resources, drive economic growth and strengthen the U.S. AI ecosystem's competitive edge.
- **Digital Transformation:** The U.S. government must modernize its public sector IT infrastructure to align with the private sector's pace of AI adoption, as current computer networks and devices remain unprepared for the necessary technology changes. The United States currently spends approximately $100 billion annually maintaining antiquated legacy systems relied upon by public agencies, which underscores the urgent need for an upgrade. Modern mobile communications networks are critical to support the widespread deployment of AI across government operations, as they are central in enabling data transmission and real-time decision-making essential for effective AI implementation.

Policy Recommendations:

- **The Trump administration should invest in Modern Infrastructure by** prioritizing digital transformation through strategic investments in modern PCs, software and all technology across government agencies to ensure that the U.S. government evolves fast or faster than the private sector.
- **The Trump administration should continue to prioritize investment in a domestic 5G ecosystem** and expedite "rip and replace" programs for national 5G networks to enhance both the operational speed and security of U.S. networks.
- **The Trump administration should establish a National AI Compute Strategy** to expand AI supercomputing capacity, leveraging existing government and academic research infrastructure.
- **The Trump administration should incentivize the acceleration of domestic AI chip production**, reducing reliance on foreign supply chains.
- **The Trump administration should enable AI data center optimization** by supporting incentives and funding for advanced cooling solutions, high-density data center designs, and energy-efficient systems. This support can complement initiatives like the Stargate AI project, which aims to build next-generation data centers and ensure the U.S. leads in scalable, efficient AI infrastructure.

2. Ensuring Energy Resilience and Infrastructure Modernization for Sustainable AI Growth

The rapid proliferation of AI-driven applications—such as machine learning models, generative AI tools and large-scale data processing—has ushered in an era of unprecedented computational demand. These energy-intensive technologies are transforming industries like healthcare, finance, entertainment and logistics, but they also place significant strain on the U.S. power grid. Data centers, which house the servers powering these applications, are projected to account for a growing share of national electricity consumption. McKinsey & Company projects that demand for AI-ready data center capacity will rise at an average rate of 33% annually between 2023 and 2030. By 2030, around 70% of total demand for data center capacity will be for centers equipped to host advanced AI workloads.

This surge in demand coincides with existing challenges in states like Texas and California, where energy infrastructure already grapples with high demand, seasonal fluctuations and the integration of renewable energy sources.

Without proactive investment in grid modernization and future-ready energy solutions, these constraints—compounded by regulatory hurdles and the pace of infrastructure upgrades—could hinder AI expansion, forcing organizations to rethink deployment strategies or investments in energy-efficient solutions. As global competition intensifies, the U.S. must act decisively to bolster its energy infrastructure, to ensure it can sustain AI growth while maintaining leadership in technological innovation.

Key Considerations:

- **Escalating Energy Demand for AI:** S&P Global Ratings estimates that data centers could require an additional 150-250 terawatt-hours of electricity in the U.S. between 2024 and 2030, necessitating about 50 gigawatts of new generation capacity, underscoring the urgent need for fast tracked, scalable energy solutions.
- **Energy Efficiency:** While modern servers have achieved significant improvements in energy efficiency compared to their predecessors, with efficiency enhancements ranging from 50% to 300%, the sheer scale of AI data-processing and inferencing workloads necessitates strategic investments in diversified, reliable and dispatchable energy sources.
- **Resilient and Modernized Grid Infrastructure:** To support widespread AI adoption, data centers must be co-located with energy infrastructure projects, and the national grid must be upgraded to handle increased loads and integrate diverse energy sources effectively.

Policy Recommendations:

- **The Trump administration should strengthen Grid Resilience and Modernization** by expanding investments in national transmission infrastructure and deploying AI-powered smart grids to optimize energy consumption, enhance grid efficiency, and ensure AI-intensive workloads do not overwhelm the existing power supply, particularly in high-demand regions.
- **The Trump administration should support Diversified Energy Sources for AI Infrastructure** by increasing federal support for a range of diversified and dispatchable energy solutions—including small modular reactors (SMRs), fusion reactors, solar, hydropower, and other power sources—tailored to AI compute growth, ensuring reliable and sustainable energy access for data centers.
- **The Trump administration should promote Energy-Efficient Data Centers and Streamlined Permitting** incentivizing the adoption of advanced technologies like liquid cooling, digital twin systems, and waste heat recovery to minimize energy consumption in AI data centers. Additionally, it should streamline permitting by creating pragmatic, expedited pathways for both public and private energy-efficient data centers, upgrading of existing data centers, power generation plants, transmission infrastructure, and renewable energy projects.
- **The Trump administration should establish an AI Infrastructure Working Group,** comprising of a dedicated task force to guide AI-driven energy policy and investment, fast-track major enterprise AI programs, and ensure alignment between technological growth, energy resilience strategies, and infrastructure development.

Innovate: Driving AI Leadership Through Public-Private Collaboration

3. Strengthening AI Workforce Development

A lack of AI talent threatens U.S. competitiveness in AI innovation, as the nation aims to keep pace with the demands of a rapidly evolving technological landscape. The advancement and deployment of AI technologies—spanning autonomous systems, agentic, natural language processing and predictive analytics—require a workforce proficient in specialized skills such as machine learning, data science, and software engineering. Yet the U.S. is facing talent shortages in these areas due to insufficient training and educational pipelines. The U.S. must develop a national AI workforce strategy to ensure a steady supply of skilled professionals, incorporating initiatives like expanded STEM education, public-private partnerships and reskilling programs to prepare both new entrants and existing workers for an AI-driven economy. Without such a strategy, industries critical to national security and economic growth—including defense, healthcare and manufacturing—risk falling behind, undermining the U.S.'s position as a global leader in technology. Given the escalating pace of AI breakthroughs and the strategic importance of maintaining a competitive edge, now is the pivotal time for the U.S. administration to embed a robust workforce development framework within the AI Action Plan to secure the talent pipeline essential for future innovation and resilience.

Key Considerations:

- **AI & STEM Talent Gaps:** The World Economic Forum predicts a 40% growth in AI and machine learning specialists by 2027.
- **Workforce & Industry Needs:** While agentic AI architecture will reduce the barriers for AI use, accelerating the adoption of AI will still require a robust workforce trained in data science, machine learning and automation.
- **K-12 AI Education:** AI literacy must start early to ensure a strong future talent pipeline.

Policy Recommendations:

- **The Trump administration should develop a national AI Workforce Strategy** by launching a multi-agency task force to coordinate federal efforts in addressing AI talent shortages. This task force should prioritize creating a centralized framework for tracking AI workforce needs, identifying critical skill gaps across industries, and aligning federal resources to support targeted training initiatives.
- **The Trump administration should empower states to integrate AI curricula in K-12 education** and provide incentives for schools adopting AI-focused programs. The administration should provide updated guidance on how AI solutions that enable education comply with federal data privacy laws for minors (FERPA and COPPA).
- **The Trump administration should establish AI skilling and certification programs** for government employees to support workforce transformation, to include broad education for the federal workforce to develop AI literacy and support the uptake and opportunities that AI brings.
- **The Trump administration should incentivize the development and use of AI tools in the trade sector**, encouraging apprenticeship programs and trade schools to integrate these tools early in their curriculum.
- **The Trump administration should modernize the H-1B visa program** to attract and retain top international AI talent.
- **The Trump administration should bolster R&D funding and collaboration for AI innovation** by establishing a multi-year AI Research Fund to ensure stable funding and enhance public-private partnerships. This fund should expand cooperative research

agreements, encourage private sector investment to accelerate AI innovation, and include incentives for private companies to engage with their communities to build AI skills.

4. Unlocking AI for Government Efficiency

AI can be leveraged by the Trump administration and its agencies to increase government efficiency and effectiveness while decreasing costs for taxpayers and meeting the evolving needs of our dynamic nation. As federal agencies manage vast amounts of data and complex operations, AI has the potential to significantly enhance government operations, leading to substantial cost savings and improved public services that directly benefit citizens. AI-powered automation in federal agencies can optimize administrative processes by streamlining and reducing processing times to improve decision-making through predictive analytics that inform policy and resource allocation.

Key Considerations:

- **AI-Powered Government Modernization:** AI and data-driven decision-making will reshape how governments operate and deliver essential services.
- **Cloud & On-Premises AI Deployment:** Balancing cloud and on-premises AI implementation can help reduce costs and enhance security. Not selecting the most appropriate infrastructure for AI programs may result in significant future cost and risk challenges.

Policy Recommendations:

- **The Trump administration should conduct a comprehensive audit of all existing federal IT systems and networks** to assess AI readiness, identify capability gaps, and evaluate operational needs.
- **The Trump administration should adopt a Cloud Smart Strategy** that optimizes the use of public, private and hybrid cloud providers to enhance data control and security, access top-tier services, offer workload optimization flexibility, and manage costs effectively.
- **The Trump administration should streamline AI technology adoption** by reforming bureaucratic barriers, accelerating procurement processes, and allocating regular funding for tech modernization. Additionally, the administration should adopt a cloud assessment model to determine the best infrastructure based on service needs, integrate advanced security approaches into the procurement requirements, and leverage technology solutions with GSA to automate government processes to support paperless initiatives, improve efficiency and reduce costs.

Evolve: Agile AI Policy & Future-Ready Governance

5. Crafting Agile AI Governance for Growth

Policy parameters within AI should be agile from the outset and allow for regular realignment and evaluation of rules to keep pace with the technology's burgeoning growth and progress, ensuring the nation remains at the forefront of global innovation. Policymakers should seek not to regulate the technology itself, but the intent of its applications, fostering an environment where innovation thrives alongside trust and responsibility. As AI continues to evolve rapidly—driving advancements in fields like healthcare, transportation, and education—the U.S. government must adopt a flexible regulatory framework that can adapt to emerging capabilities, ethical considerations, and global economic opportunities while encouraging open collaboration and responsible development. This approach should be applied across the spectrum of the U.S. government's governance and security structures, from economic agencies to security departments, to ensure that critical assets and innovation are protected as integral facets of our economy, supporting both private-sector growth and public-sector resilience.

Key Considerations

- **Rapid Technological Evolution:** AI's fast-paced development requires policies that can adapt to new use cases and capabilities, such as generative AI and autonomous systems, without stifling innovation. PwC research found that 76% of executives believe AI introduces new risks that require regulatory attention, and many expressed the need for clearer guidelines to manage those risks effectively, during this time of technological innovation.
- **Ethical Governance:** Balancing innovation with ethical oversight is essential to maintain public trust and ensure AI serves societal benefits across healthcare, education and beyond. Dell research found that 83% of business leaders see AI regulations as important to maximize the potential of AI for generations to come, suggesting that thoughtful governance can enhance adoption rather than hinder it.
- **Economic Competitiveness:** Agile policies can protect U.S. intellectual property, champion U.S. innovation, incentivize domestic R&D, and support open ecosystems to sustain global economic advancement in AI.

Policy Recommendations

- **The Trump administration should foster a pro-innovation regulatory sandbox**, allowing businesses and academia to test AI solutions in a controlled environment, accelerating development and deployment across key industries.
- **The Trump administration should implement intent-based AI guidelines** that focus on the purpose and outcomes of AI applications, rather than rigid technical restrictions, to encourage innovation while addressing risks.
- **The Trump administration should support broad, open AI ecosystems** built on trust and transparency by promoting open-source model development, ensuring developers adhere to industry standards for responsible AI while maintaining controlled releases to mitigate risks of data misuse and safeguard national interests.
- **The Trump administration should establish an AI Policy Task Force**, comprising of public and private sector leaders from the AI industry to regularly review and update regulatory frameworks, ensuring they remain flexible, aligned with technological advancements, and supportive of both proprietary and open-source innovation to meet national priorities.
- **The Trump administration should establish a Domestic AI Competitiveness Initiative** aimed at championing and safeguarding American excellence in AI innovation, navigating evolving international policies and incentivizing domestic R&D to sustain U.S. technological sovereignty and global market leadership.
- **The Trump administration should establish a robust international forum** led by the U.S to convene regular engagements with key global partners, to mitigate regulatory burdens that hinder domestic U.S. tech companies operating within the global AI ecosystem. This initiative should focus on fostering cooperative dialogue to harmonize, where appropriate, regulatory approaches, and support the competitive edge of American technology firms while promoting a balanced, collaborative and international approach.

6. Enhancing AI Cybersecurity

AI-driven cyber threats are increasing in scale and complexity, necessitating AI-powered security and resiliency solutions to safeguard critical infrastructure and national security. As reliance on digital systems grow—spanning energy grids, financial networks and transportation—AI can enhance protection by detecting anomalies, predicting risks, and improving response times, ensuring the resilience of vital assets.

Key Considerations:

- **AI-Powered Cyberattacks:** Malicious actors increasingly deploy AI-driven cyber threats that require automated defense systems.
- **Zero Trust Frameworks:** Secure AI deployment necessitates robust, layered cybersecurity architecture.
- **National Security AI Coordination:** AI must be integrated into federal cybersecurity frameworks.

Policy Recommendations:

- **The Trump administration should seek to adopt autonomous AI cybersecurity solutions** for real-time threat detection, response, and adaptation to match malicious AI activity, with regular private sector engagement to assess effectiveness and industry impact.
- **The Trump administration should mandate zero-trust security standards** with response capabilities to protect AI applications and critical infrastructure, including a voluntary compliance register.
- **The Trump administration should coordinate a unified AI security framework** across federal agencies, prioritizing trusted systems, standardized protocols, and private-sector collaboration, with a dedicated task force to oversee implementation.

Conclusion: ***A Blueprint for U.S. AI Global Leadership***

The U.S. AI Action Plan represents a defining moment to advance American leadership in AI innovation on the world stage. By boldly prioritizing AI infrastructure, a skilled workforce, open AI ecosystems, and robust security, the U.S. can forge a powerful, repeatable blueprint for global leadership—one that fuses AI deployment and innovation with decisive action, transforming financial commitment into a tangible strategy.

Dell Technologies stands ready to partner with the U.S government to propel this vision into reality. We look forward to collaborating with policymakers, industry partners, and academic pioneers to architect an AI ecosystem that empowers every American, securing a future-ready infrastructure that not only sustains but amplifies U.S. global competitiveness in AI for decades to come.